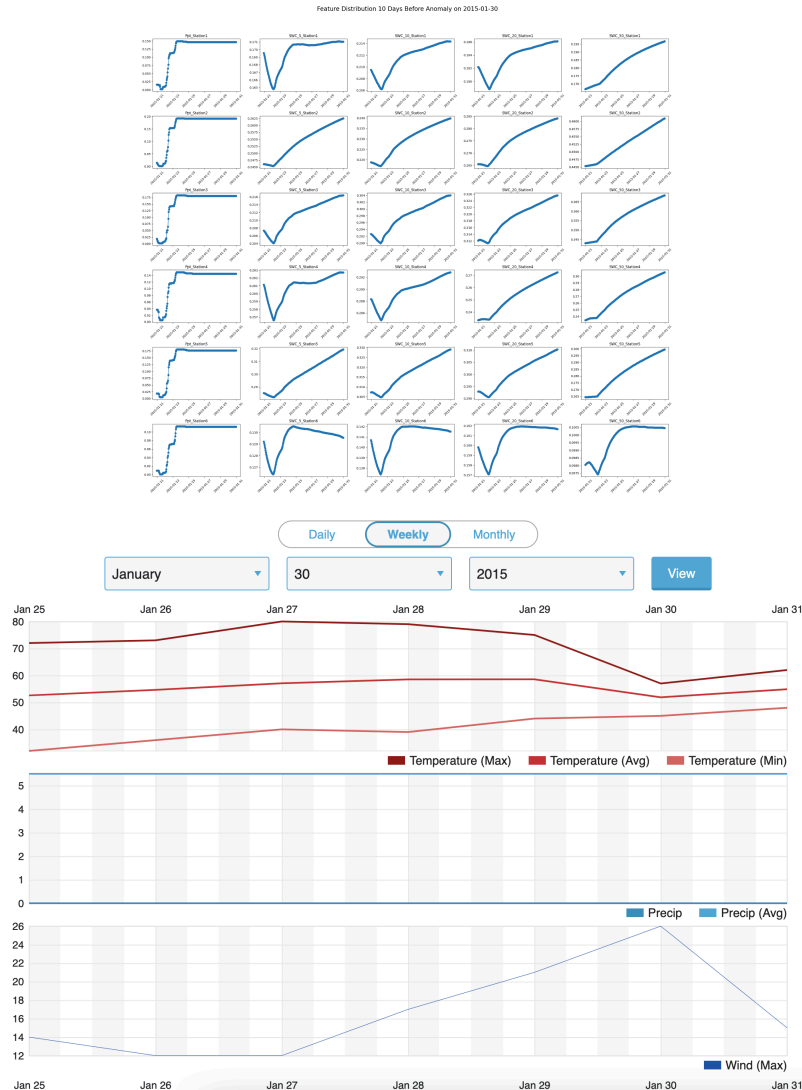
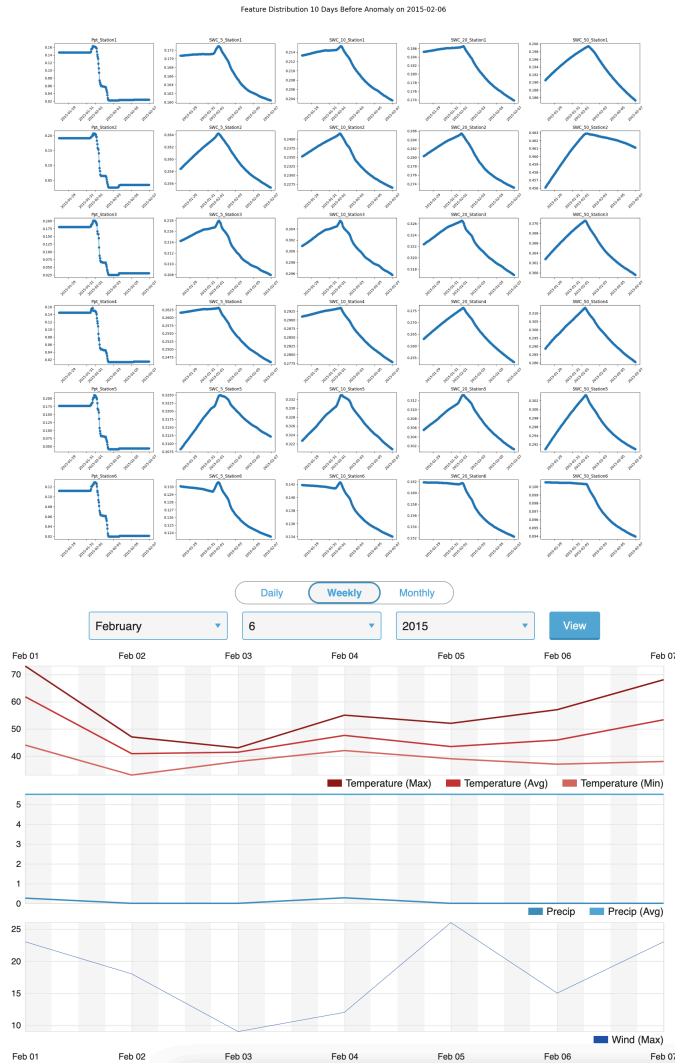


Multivariate Anomaly Detection:

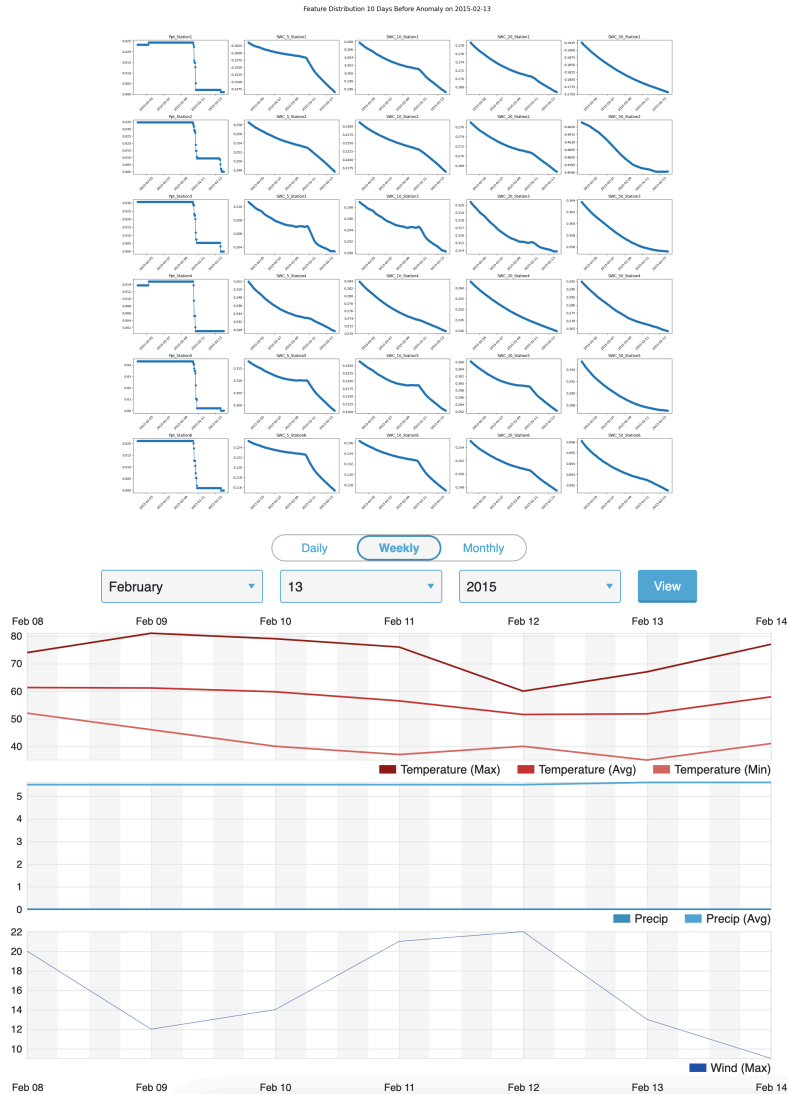
The multivariate LSTM autoencoder detected three anomalies that occurred on January 30, 2015, February 6, 2015, and February 13, 2015. The anomaly on January 30, 2015, was primarily caused by a spike in precipitation from approximately 0 to 0.15 - 0.20 mm across all six stations on January 22, 2015. This event is reflected in every other feature as the SWC_5, SWC_10, SWC_20, and SWC_50 readings all dramatically increased on January 22, 2015.



The anomaly on February 6, 2015, was primarily caused by a dropoff in precipitation values from 0.18 - 0.20 mm to 0.02 - 0.05 mm around February 1, 2015. This is likely a reversion to the mean after the high rainfall that occurred a week earlier as detailed above. This event is reflected in every other feature as the SWC_5, SWC_10, SWC_20, and SWC_50 readings all dramatically decreased around February 1, 2015.

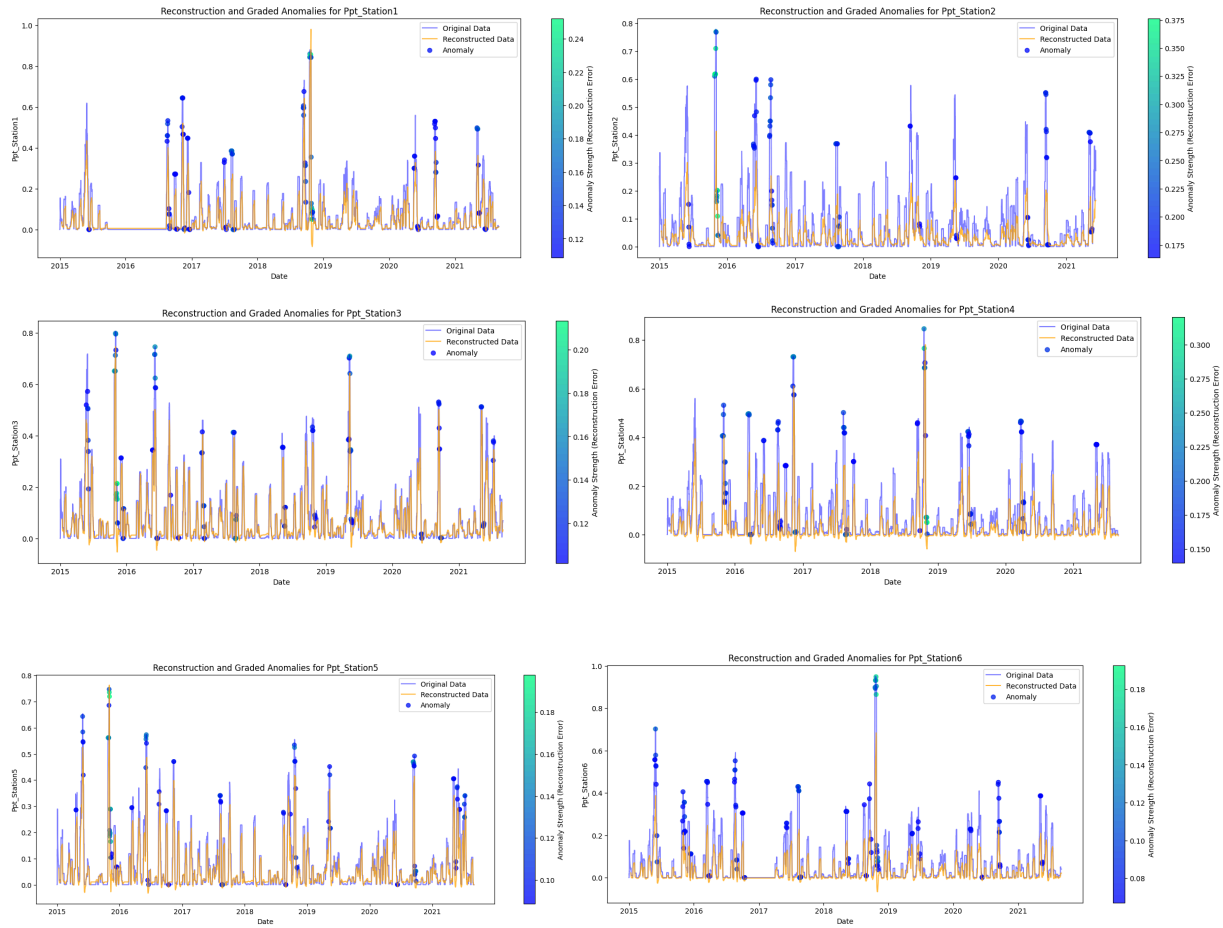


The anomaly on February 13, 2015, was primarily caused by a dropoff in precipitation values from 0.01 - 0.04 mm to 0 mm around February 11, 2015. This dropoff is likely a further reflection of the decrease in rainfall compared to the extreme precipitation in late January 2015. This event is reflected in most other features as most reading dramatically decreased around February 11, 2015, but some stations (especially Station 4) exhibit no noticeable discrepancy between the rates of decline before and after February 11, 2015.



Precipitation Anomaly Detection:

The strongest precipitation anomalies were found in two distinct periods (October 25, 2015 - November 12, 2015 and October 17, 2018 - November 3, 2018). Each period is characterized by a dramatic increase in rainfall and a subsequent dropoff. The LSTM autoencoder trained on the precipitation time series data flagged almost every day across each of these periods in both the spike and dropoff in precipitation levels.



LSTM SWC Anomaly Detection:

An LSTM model was trained to predict soil water content levels across stations and detect anomalies based on the difference between the predictions and actual data. The strongest anomalies were found in a period from June 2017 - August 2017 as well as from April 2021 - June 2021. Most of the detected anomalies were abnormal increases in SWC levels as opposed to extreme dropoffs.

